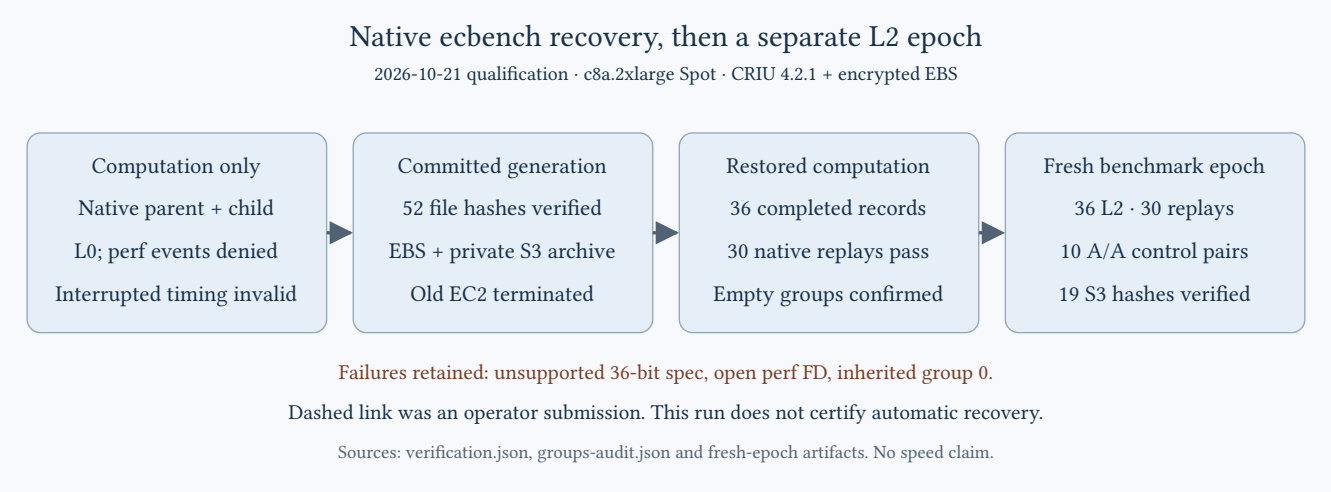
# Native ecbench CRIU / EBS qualification

Iso Lab · 21 October 2026 · operator-controlled validation

**Observed:** a native crypto process tree resumed on a replacement Spot worker and completed 36 records. Independent native replay reproduced all 30 measured records. These computations remain **L0, with invalid benchmark timing**. A distinct fresh epoch earned 36/36 L2 observations, 30/30 replays, 10 admitted A/A pairs and 19/19 verified S3 payload hashes. This run does not certify automatic recovery.




## Frozen workload and failures

The supported workload is a public 32-bit prime_search fixture (seed 59297), two targets (seed 20261002), rho.negation, rho.plain and matching A/A; five measured rounds and one warmup. The original UTC deadline was 2026-10-21 06:25. All computation and fresh-epoch execution finished before it; later evidence collection and independent replay do not represent new admitted measurements.

The first 36-bit request was rejected before a child started. A supported measured process then failed CRIU dump because it held a perf_event descriptor. Both failures are retained. A native launcher forced L0, denied performance-counter creation, and retained the original math workload. Its actual parent and active child checkpointed with CRIU 4.2.1 on AL2023 kernel 6.18.54-121.166.amzn2023.x86_64.

The first restore inherited supplementary group 0. After that copy completed and its Pod was removed, a second nonconcurrent restore cleared the supervisor's groups. Its parent and child then had UID/GID 1000, empty supplementary groups, zero capabilities, no_new_privs and seccomp. That copy also completed 36 records and passed 30 independent native replays. This establishes the tested trusted process procedure, not isolation for arbitrary untrusted images.

| Check | Recorded evidence |
|---|---|
| Original worker | `i-0ab01e249ab5475f7` c8a.2xlarge, us-east-2c |
| Replacement worker | `i-0ab0f5f4afc72d5d6` c8a.2xlarge, us-east-2a |
| Writer fence | Original EC2 confirmed terminated before replacement Pod creation |
| Committed filesystem | `snap-0ead1fd75c8add942` encrypted, complete; 52 file hashes |
| Private archive | Versioned S3 archive passed complete download / byte comparison |
| Fresh A/A | 10 pairs; recorded 95% ratio interval 0.9997380265–1.0011447800 |
| Fresh results | 36 L2 observations, 30 native replays, 19 S3 payload hashes |

## Identity, scope and evidence

Source: `cb2ec6873f30b683f7885b53c410d69965f02c20`. Native SHA-256: `583e62f0836eae43330c4b36bbe3d145dacb01de3132436c71f6924445c6a5a9`. Fresh Job: `criu-ecbench-20261021-epoch-1`. Fresh Pod UID: `848af771-ccde-46ce-aef5-7e73e2de357e`.

The fresh job used the maintained AWS/CodeBuild harness with a pinned deployed image. It was not the pending PR 2342 full main-branch commit-building Action. A transient Spot quota/capacity error cleared within the unchanged 96-vCPU account quota and 64-vCPU pool cap. No failed L2/A/A epoch was selectively retried.

Full identities, protocols, exact commands, private archive version and source hashes are in `REPORT.md`, `verification.json`, `COMMITTED.json`, `protocol-v3.json`, `groups-audit.json` and the fresh-epoch artifacts. Raw memory stays private and is absent from Git. Canonical performance graphs and leaderboards remain unchanged: this is infrastructure qualification, not an algorithm speed comparison. The diagram records execution and validation counts.